

Model Verification 4: Error Analysis

We analyze the error term (regression residual) to determine if we ignored factors driving trading cost. We compute the error term δ_i (difference between estimated and actual) as follows:

$$\delta_i = \text{Estimated MI} - \text{Actual Cost}$$

Then we regress δ_i on factors: market movement, side of order (buy vs. sell), sector, order size (to determine robustness of fit), market cap, and so forth. A statistically significant result would indicate the “factor” is a consistent contributor to trading cost.

Important Note: Analysts should perform data verification across all sample orders, grouping data into categories to determine bias. For example, you should perform data verification by order size categories, side of the order (buys and sells separate), by sector, volatility, and market movement (up days and down days). If bias is present, you need to discover where and why bias occurred and follow through to solutions.